DEPI INDUSTRY GRADUATION PROJECT — DOCUMENTATION PROCESS.

1. Project Planning & Management.

Project Title: ***"Adaptive Learning Assistant: An AI Powered Learning Chatbot for Dynamic Student Engagement".***

GIZ5_IND1_G1

Project Members:

1- Hosam Hasan Abdelhady Ibrahim. (Team Leader)

2- Omnia Reda Mohamed Rizk Ghalwash.

3- Mohamed Hussien Mohamed.

4- Doaa Abdelraouf Mahmoud Elganzoury.

1.1 Project Proposal

1.1.1 Overview

The Adaptive Learning Assistant is an intelligent, multi-modal conversational learning agent designed to elevate student engagement and streamline curriculum-focused study for High& Higher education as well as training programs. Powered by n8n workflow orchestration, Google Gemini/Claude as examples of LLM/embeddings, and a Vector Database (RAG), the system delivers dynamic, context-aware pedagogical interactions via Telegram. The assistant bridges the gap between static academic materials and interactive learning by providing automated textual explanations, Multiple-Choice Questions (MCQs), and visual concept diagrams and charts on demand.

1.1.2 Objectives

- **Automate Curriculum Ingestion:** Build an offline ingestion pipeline in n8n that automatically fetches, processes, extracts text from PDF documents, generates dense vector embeddings, and indexes content into a *Supabase Vector Store* as an example.
- **Deliver Dynamic Multi-Modal Learning:** Provide personalized study assistance through three core specialized runtime sub-agents: Text QA (RAG-backed answers), MCQ Generator (structured JSON evaluations with answer explanations), and Visualization (interactive charts via QuickChart API and custom visual aids via Imagen/DALL-E as examples).
- **Optimize Student Engagement & Access:** Deliver a highly responsive, intuitive interface through Telegram, eliminating user friction while maintaining context-aware conversational memory.
- **Ensure Strict Domain Guardrails:** Implement an automated scope and intent classifier at the runtime pipeline to restrict chatbot interactions strictly to the selected course material as a beginning, rejecting out-of-scope inquiries and unsupported file uploads.

1.1.3 Project Scope

In-Scope Deliverables:

- Subject-specific knowledge boundaries.
- Runtime Student Chat Assistant Pipeline.
- Question answering.
- Multiple-Choice Questions (MCQs) generating, with answers and explanation.
- Visual concept diagrams and charts as illustration tools.
- Scope & Intent Classifier (Gemini Chat + Memory Tool + Prompt Template).
- Branching routes process (TEXT_QA, MCQ_GEN, IMAGE_GEN, OUT_OF_SCOPE).
- Integration of external APIs.
- Telegram Bot end-user interface setup and prompt engineering optimization.
- Comprehensive testing process.

Out-of-Scope / Exclusions:

- Acceptance or processing of direct file/photo/video uploads from students during runtime chat sessions (enforced automatic rejection policy).
- Answering general, non-curriculum, off-topic, or outside knowledge questions beyond indexed course vector stores.

1.2 Project Plan & Schedule

Project Milestone / Item	DEPI Target Deadline	Key Deliverables & Output
Project Planning & Management	**2 Oct 2026**	Project charter, Gantt chart schedule, role allocations, risk matrix, KPI metrics definition, GitHub repo setup.
Literature Review	**9 Oct 2026**	Comparative study on RAG architectures, n8n orchestration efficiency, embedding accuracy, and AI pedagogy guardrails.
Requirements Gathering	**9 Oct 2026**	Functional & non-functional requirements specification, n8n workflow node mappings, API quota & memory specs.
System Analysis & Design	**9 Oct 2026**	Architecture diagrams, sequence

		workflows, Supabase schema design, system guardrail & security policies.
Implementation (Source Code & Execution)	**16 Oct 2026**	Fully functional n8n Workflows 1 & 2, Supabase vector indexing, Gemini & QuickChart API integrations, bot testing (as examples).
Final Presentation & Testing & Reports	**16 Oct 2026**	Comprehensive evaluation report, KPI validation results, final presentation deck, live system demo, full GitHub commit.

1.2.1 Project Schedule Breakdown (Gantt Chart Timeline)

Phase / Module	Start Date	End Date	Dependencies
Phase 1: Project Setup & Management	24 Sep 2026	02 Oct 2026	None
Phase 2: Requirements & System Design	02 Oct 2026	09 Oct 2026	Phase 1
Phase 3: Literature & Tech Stack Research	02 Oct 2026	09 Oct 2026	Phase 1
Phase 4: Workflow 1 (Course Ingestion)	05 Oct 2026	12 Oct 2026	Phase 2
Phase 5: Workflow 2 (Runtime Assistant)	08 Oct 2026	15 Oct 2026	Phase 4
Phase 6: Integration & Guardrails Testing	12 Oct 2026	15 Oct 2026	Phase 5
Phase 7: Final Documentation & Presentation	14 Oct 2026	16 Oct 2026	Phase 6

1.3 Task Assignment & Team Roles

To ensure streamlined execution, responsibilities are distributed across specialized engineering roles covering workflow automation, vector database administration, AI prompt engineering, and quality assurance.

Role Title	Primary Responsibilities	Assigned n8n / System Components
Project & System Architect. *(Hosam & Omnia)*	Overall project management, timeline tracking, system architecture design, GitHub repo maintenance, final reporting.	Whole System Architecture, Project Management Docs, Final Presentation.
AI & Workflow Course Ingestion Engineer. *(Hosam, Omnia, Mohamed & Doaa)*	Building and configuring Workflow 1: HTTP binary fetch, PDF text extraction, Google Gemini Embeddings integration for example, Supabase Vector indexing.	Workflow 1 (Manual Trigger, HTTP Request, Read Binary, Extract PDF, Embeddings, Supabase Indexing).
AI & Conversational Bot Engineer. *(Hosam, Omnia, Mohamed & Doaa)*	Building Workflow 2: Telegram Bot triggers, Scope & Intent Classifier, Prompt Engineering for Memory & Guardrails, Sub-agent routing.	Workflow 2 (Telegram Trigger, Scope & Intent Classifier, Chat Model, Memory Tool, Prompt Template).
Multi-Modal Integration & Agent Specialist. *(Hosam, Omnia, Mohamed & Doaa)*	Configuring sub-agent execution routes: RAG Text QA, MCQ JSON Generator, API integration, Imagen/DALL-E visual generation for example.	Sub-Agents (Supabase Vector Store, MCQ Generator, QuickChart API, Imagen/DALL-E Image Gen).
Quality Assurance & Guardrail Tester. *(Mohamed & Doaa)*	Testing guardrails (file upload auto-rejections, off-scope handling), response latency benchmark, SQL log auditing, KPI verification.	Guardrail Testing, System Policy Enforcement, Bug Tracking & GitHub Issue Resolution.

1.4 Risk Assessment & Mitigation Plan

A proactive risk management framework has been established to identify technical, architectural, and operational risks associated with building an n8n-orchestrated AI chatbot.

Risk Identifier & Category	Description	Likelihood / Impact	Mitigation Strategy	Contingency Plan
R-01: System Scope Breach	Students submitting off-topic or general knowledge queries causing hallucinated or unverified responses.	**Medium / High**	Implement rigid Scope & Intent Classifier with strict system prompts blocking outside knowledge.	Route all unverified intents directly to the 'OUT_OF_SCOPE'.
R-02: Unauthorized File Uploads	Students uploading large PDFs, images, or executable files into the runtime Telegram chat.	**High / Medium**	Enforce IF Node rule: check if document/photo/video exists; if TRUE, trigger auto-rejection instantly.	Hard-stop execution using the 'Stop Execution' node and return clear policy warning.
R-03: API Rate Limits & Quotas	Exceeding Google Gemini, Supabase, or QuickChart API key rate limits during peak testing.	**Medium / High**	Implement exponential backoff retry policies in n8n HTTP nodes and cache frequent query responses.	Maintain secondary fallback API keys or local vector store mirror for testing continuity.
R-04: Heavy PDF Parsing Failure	Complex, multi-column, or scanned PDF course materials failing during text extraction.	**Low / Medium**	Pre-process and validate PDF binary format before ingestion; use robust recursive character splitters.	Fallback to OCR-based pre-processing or manually verified clean text files before embedding.
R-05: High Response Latency	Multi-stage RAG retrieval + LLM synthesis taking >5 seconds, degrading user experience.	**Medium / Medium**	Optimize chunk size and overlap parameters in Supabase Vector Store; stream responses.	Send immediate Telegram typing status indicator to keep student informed while processing.

1.5 Key Performance Indicators (KPIs)

Project success will be evaluated across technical efficiency, system security/guardrail adherence, and educational usability metrics:

KPI Category	Performance Metric	Target Benchmark	Measurement Method
System Performance	Average Query Response Time (End-to-End)	**< 5.0 Seconds**	n8n execution time logs & Telegram response timestamp tracking
System Performance	System Availability & Uptime	**99.5% Uptime**	n8n webhook listener uptime and Supabase service status monitoring
Guardrails & Security	Out-of-Scope Rejection Accuracy	**100% Accuracy**	Automated test suite submitting 50 non-curriculum queries
Guardrails & Security	File Upload Interception Rate	**100% Rejection**	Testing direct uploads of PDF, JPG, and MP4 files via Telegram
Educational Accuracy	RAG Retrieval Context Relevance	**> 92% Accuracy**	Expert architectural/curriculum manual review against source text
Educational Accuracy	MCQ Generation JSON Validity	**100% Valid JSON**	Automated JSON schema validation in n8n MCQ Generator Agent
User Adoption	Student Engagement Rate	**> 80% Active Rate**	Telegram bot active chat session analytics during user evaluation
User Adoption	User Satisfaction Score (CSAT)	**> 4.0 / 5.0**	Post-interaction 1-click survey feedback delivered via Telegram

1.6 Technical Annex: n8n Workflow Architecture Mapping

This section outlines the primary chosen workflow plan; however, individual workflow elements and nodes remain subject to refinement during the implementation and testing phases.

The project execution relies directly on two synchronized n8n workflows as detailed in the technical specification architecture:

Workflow 1: Curriculum Data Ingestion (Offline Pipeline)

- Manual Trigger -> Initiates the batch vectorization process.
- HTTP Request (Get PDF URL) & Read Binary File (Load PDF) -> Fetches academic course materials.
- Extract PDF Text -> Parses raw text from PDF binary stream.
- Google Gemini Embeddings (text-embedding-004) & Supabase Vector Store -> Generates dense vector representations and indexes document chunks into vector database.
- SQL Database Integration & Update Status (Complete) -> Logs indexing metadata and execution status.

Workflow 2: Student Chat Assistant (Runtime Pipeline)

- Telegram Trigger (Listen for message) -> Captures student queries in real time.
- IF Node Guardrail (Check for File Uploads) -> IF document/photo/video exists = TRUE -> Route to Telegram Send Rejection & Stop Execution; IF FALSE -> Route to Scope & Intent Classifier.
- Google Gemini Chat (Scope & Intent Classifier) -> Equipped with Chat Model, Memory Tool, and Prompt Template; outputs structured JSON with identified intent and clean query.
- Branch 1: TEXT_QA -> Supabase Vector Store (Retrieve Context) -> Google Gemini Chat (RAG Model) -> Telegram (Send Text).
- Branch 2: MCQ_GEN -> AI Agent (MCQ Generator) -> Structured JSON & Explanations -> Telegram (Send MCQ Text).
- Branch 3: IMAGE_GEN -> AI Agent (Visualization) -> Dual outputs to QuickChart API (charts) & Imagen/DALL-E (visual concepts) -> Telegram (Send Photo/Image).
- Branch 4: OUT_OF_SCOPE -> Telegram (Send Out of Scope Rejection message).